ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PRACTICE EXAM 1

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 8 strands: Life and Living (photosynthesis and respiration, food webs, micro-organisms), Matter and Materials (particle model, density, atoms, elements and compounds, chemical reactions), Energy and Change (static electricity, series and parallel circuits, visible light) and Planet Earth and Beyond (the solar system, galaxies and space exploration).

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL calculations with formulae and units. A non-programmable calculator may be used.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Plants take in which gas for photosynthesis?

A Oxygen B Carbon dioxide C Nitrogen D Hydrogen

1.2 Which of these is a product of respiration?

A Glucose B Oxygen C Chlorophyll D Carbon dioxide

1.3 Organisms that break down dead plants and animals are called …

A producers. B decomposers. C consumers. D predators.

1.4 Which disease is caused by a virus?

A Tuberculosis B Cholera C Influenza D Athlete's foot

1.5 The smallest particle of an element that still has the properties of that element is …

A an atom. B a compound. C a mixture. D a cell.

1.6 Water (H_2O) is …

A an element. B a compound. C a mixture. D a metal.

1.7 When a substance is heated, its particles …

A become bigger. B stop moving.
C disappear. D move faster and further apart.

1.8 When a balloon is rubbed on dry hair, the balloon becomes …

A electrically charged. B magnetic.
C hotter than boiling water. D a conductor.

1.9 The splitting of white light into its colours by a prism is called …

A reflection. B absorption. C dispersion. D diffusion.

1.10 Which planet is closest to the Sun?

A Venus B Earth C Mercury D Mars

(10)

QUESTION 2: SCIENTIFIC TERMS

Give the correct scientific term for each description.

2.1 The green pigment in leaves that absorbs light energy

2.2 Many food chains linked together in an ecosystem

2.3 All the living and non-living things in an area and how they interact

2.4 Living things so small that they can only be seen with a microscope

2.5 A substance made of two or more different elements that are chemically joined

2.6 The spreading of particles from an area of high concentration to an area of low concentration

2.7 The mass of a substance per unit volume

2.8 A circuit in which components are connected one after another in a single path

2.9 The bending of light when it passes from one medium into another

2.10 The distance that light travels in one year

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 Write the word equation for photosynthesis. Show what is needed above the arrow. (3)

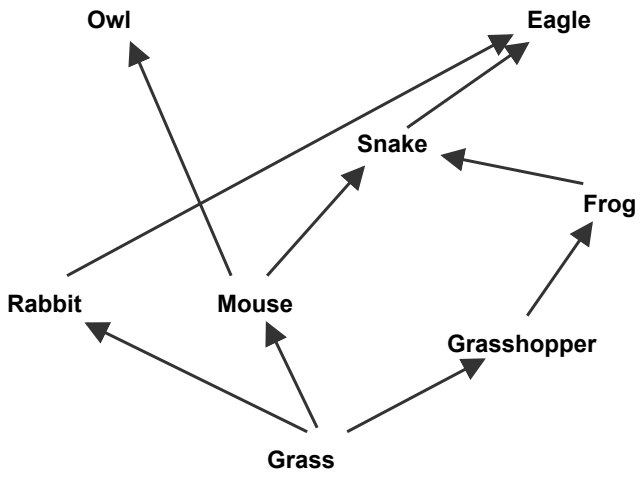
3.2 A learner tested a leaf for starch.

3.2.1 Why is the leaf placed in boiling alcohol during the test? (1)

3.2.2 What colour does iodine solution turn if starch is present? (1)

3.2.3 A pot plant was kept in a dark cupboard for 48 hours before the test. Predict the result and explain it. (2)

3.3 Study the food web.



The arrows point from the organism that is eaten to the organism that eats it.

3.3.1 Name the producer in this food web. (1)

3.3.2 Name TWO herbivores. (2)

3.3.3 Write down a food chain from this food web that contains FOUR organisms. (2)

3.3.4 What would happen to the number of frogs if all the snakes were removed? Explain. (2)

3.3.5 What is the original source of energy for this food web? (1)

3.4 (a) Give TWO ways in which micro-organisms are useful to people. (b) Explain how a vaccine protects a person against a disease. (4)

3.5 Give ONE difference between photosynthesis and respiration. (1)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 Use the particle model to describe the arrangement and movement of the particles in (a) a solid and (b) a gas. (4)

4.2 A tight metal lid on a glass jar can be loosened by holding it under hot water. Explain why. (2)

4.3 Explain why you can smell perfume from across a room a few minutes after someone opens the bottle. (2)

4.4 A block of metal has a mass of 540 g and a volume of 200 cm^3. (a) Calculate its density. Use: density = mass ÷ volume. (b) Water has a density of 1 g/cm^3. Will the block float or sink in water? (3)

4.5 Classify each substance as an element or a compound: (a) O_2; (b) CO_2; (c) Fe; (d) NaCl. (4)

4.6 How many atoms of each element are there in one molecule of sulfuric acid, H_2SO_4? (2)

4.7 When iron is left outside in damp air, it reacts with oxygen to form iron oxide (rust). (a) Name the reactants. (b) Name the product. (c) Give ONE sign that a chemical reaction has taken place. (3)

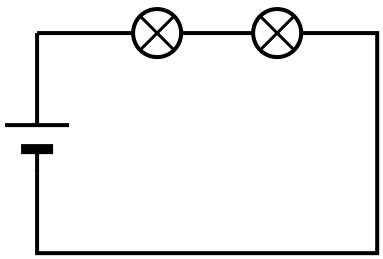
[20]

QUESTION 5: ENERGY AND CHANGE

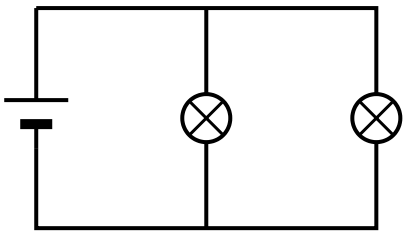
5.1 After a balloon is rubbed on dry hair, the hair stands up and is attracted to the balloon. Explain this in terms of electrons and charges. (3)

5.2 Give TWO safety rules to follow during a thunderstorm. (2)

5.3 Study the two circuits. All the cells and bulbs are identical.



Circuit A



Circuit B

5.3.1 Which circuit is a parallel circuit? (1)

5.3.2 In which circuit will the bulbs glow more brightly? (1)

5.3.3 One bulb is removed from each circuit. What happens to the other bulb in each circuit? (2)

5.3.4 Which type of circuit is used for the lights in a house? Give a reason. (2)

5.4 (a) What happens to the current in a series circuit when more resistors are added? (b) Name ONE household appliance that uses the heating effect of an electric current. (2)

5.5 Visible light

5.5.1 Name the colours of the visible spectrum in the correct order. (2)

5.5.2 Explain why a ripe tomato looks red in white light. (2)

5.5.3 State the law of reflection. (1)

5.5.4 A straw standing in a glass of water looks bent. Name and explain this effect. (2)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 List the eight planets of our solar system in order, starting with the one closest to the Sun. (4)

6.2 Give TWO differences between the inner planets and the outer planets. (2)

6.3 Pluto was called a planet until 2006. What is it called now, and why? (2)

6.4 Where is the main asteroid belt found? (1)

6.5 Describe what a comet is made of and which way its tail points. (2)

6.6 (a) What is the name of our galaxy? (b) What shape is it? (2)

6.7 The nearest star to the Sun, Proxima Centauri, is about 4,2 light-years away. How long does its light take to reach the Earth? (1)

6.8 Give TWO ways in which artificial satellites are useful to people. (2)

6.9 (a) In which South African province is part of the Square Kilometre Array (SKA) being built? (b) What type of waves from space does it detect? (2)

6.10 Who was the first person to walk on the Moon, and in which year? (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

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END OF QUESTION PAPER